



Cambridge IGCSE™

MATHEMATICS

0580/23

Paper 2 (Extended)

May/June 2021

MARK SCHEME

Maximum Mark: 70

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **6** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

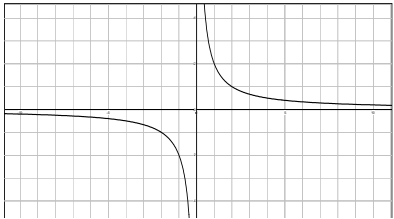
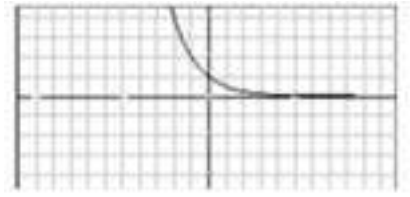
GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Maths-Specific Marking Principles	
1	Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
2	Unless specified in the question, answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
3	Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
4	Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
5	Where a candidate has misread a number in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 mark for the misread.
6	Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Question	Answer	Marks	Partial Marks
1	-24.6	1	
2(a)	$\frac{18}{25}$ cao	1	
2(b)	$\frac{1}{250}$ cao	1	
3	Corresponding	1	
4	130	2	M1 for $360 - 100$ or better
5	63	2	M1 for $600 \times \frac{1.5}{100}$ oe or better If 0 scored SC1 for answer 663
6	$100y - np$	2	B1 for $100y$ seen or for answer $[10^k]y - np$
7(a)	125	1	
7(b)	29	1	
8	162.07 cao	2	M1 for $190 \div 1.1723$
9	$\frac{5}{3} \times \frac{2}{15}$ oe or $\frac{10}{6} \div \frac{45}{6}$ oe with common denominator	M2	B1 for $\frac{5}{3}$ oe or $\frac{15}{2}$ oe or M1 for $their \frac{5}{3} \times their \frac{2}{15}$
	$\frac{2}{9}$ cao	A1	
10(a)	Translation $\begin{pmatrix} -1 \\ -8 \end{pmatrix}$	2	B1 for each
10(b)	Image at $(-1, -1)$, $(-4, -1)$, $(-1, -2)$	2	B1 for image correct scale factor and orientation but wrong position or for enlargement scale factor $\frac{1}{2}$ centre $(0, 0)$
11	$12x^7$ final answer	2	B1 for $12x^j$ or kx^7 ($j, k \neq 0$) as final answer
12	-1, 0, 1 final answer	2	B1 for $-1 \leq x < 2$ or two correct answers and no extras or three correct answers and one extra/wrong
13	$4t$ final answer	2	B1 for $6t - 6q$ or $-2t + 6q$ or $2t - 6q$ or for $4t$ or $0q$ in the final answer

Question	Answer	Marks	Partial Marks
14	$[\pm] 21$	3	M2 for $29^2 - 20^2$ oe or better or M1 for $20^2 + k^2 = 29^2$ oe
15(a)	Similar	1	
15(b)	4	2	M1 for $\frac{12}{6} = \frac{8}{BX}$ oe or better If 0 scored SC1 for answer 3.5
15(c)(i)	6.7265 or 6.73 or 6.726 to 6.727	2	M1 for scale factor 2^2 or $\left(\frac{1}{2}\right)^2$ oe soi
15(c)(ii)	13.453 or 13.5 or 13.45 to 13.46	1	FT <i>their (c)(i) × 2</i>
16	477	2	M1 for $80 - 0.5$ oe or better seen
17	72	2	M1 for $\frac{360}{180-175}$ oe or $\frac{180(n-2)}{n} = 175$ oe
18(a)	12	1	
18(b)	144	2	FT $12 \times \text{their } V$ M1 for any relevant area FT <i>their V</i>
19(a)	80	2	B1 for angle $PQT = 50$
19(b)	$[w =] 68$ $[x =] 36$	3	B1 for 68 B2 for 36 or M1 for $3x + 2x + 68 + 112 = 360$ or better
20	2.31×10^p	2	B1 for $21 \times 10^{p-1}$ or 0.21×10^p or answer with figs 231
21	14.1 or 14.12...	3	M2 for $\sin 65 = \frac{12.8}{BC}$ oe or better or M1 for recognition that the line from B is perpendicular to AC
22	$\frac{81}{(y-2)^2}$ final answer	2	M1 for $z = \frac{k}{(y-2)^2}$ oe or better

Question	Answer	Marks	Partial Marks
23	70.5 or 70.52 to 70.53	4	B3 for 59(.0) or 58.99... or 50.5 or 50.47 to 50.48 OR M2 for $\frac{10^2 + 9^2 - 11^2}{2 \times 10 \times 9}$ oe or equivalent expression for smaller angle or M1 for $11^2 = 10^2 + 9^2 - 2 \times 10 \times 9 \cos(\dots)$ oe or equivalent expression for smaller angle A1 for $\frac{1}{3}$ oe
24(a)	Correct sketch 	2	B1 for one correct branch or attempt at correct shape
24(b)	Correct sketch 	2	B1 for correct shape but crossing x-axis or for correct shape but just drawn in one quadrant
25	0 and 4 final answer	4	B3 for $5x^3(x - 4)$ or better or B2 for $5x^4 - 20x^3$ or B1 for $5x^4$ or $-20x^3$
26	0.845 oe	3	M2 for $0.7 \times 0.95 + (1 - 0.7) \times 0.6$ oe or M1 for one of these products